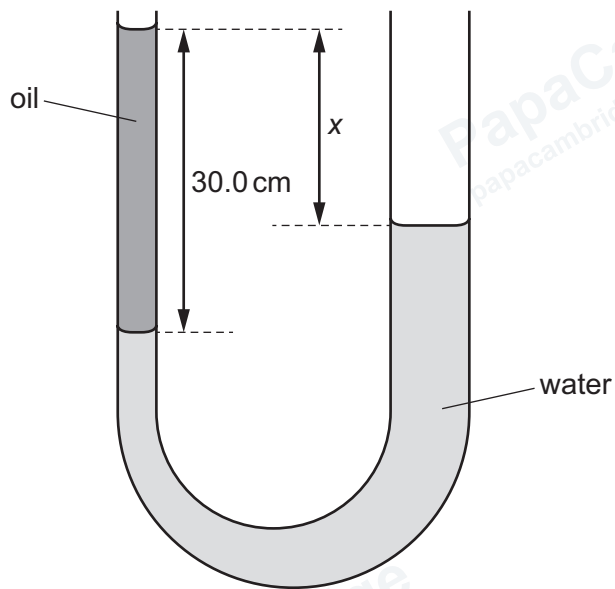


- 14 A U-tube has a left arm of cross-sectional area A and a right arm of cross-sectional area $4A$. The tube contains water of density 1000 kg m^{-3} and oil of density 850 kg m^{-3} as shown.



The column of oil on top of the water in the left-hand arm is of length 30.0 cm .

What is the difference in height x between the levels in the two arms of the tube?

- A** 4.5 cm **B** 6.2 cm **C** 23.8 cm **D** 25.5 cm